



**BHASKARACHARYA NATIONAL INSTITUTE FOR SPACE
APPLICATIONS AND GEO-INFORMATICS**

WEEKLY PROGRESS REPORT (07/09/2026 – 13/09/2026)

WEEK 7

PROJECT NAME

**CHATBOT FOR STRUCTURED, UNSTRUCTURED
AND SEMANTIC KNOWLEDGE QUERY**

PROJECT DESCRIPTION :

DESIGN AND IMPLEMENT A CHATBOT THAT CAN ANSWER TO DIFFERENT TYPE OF QUERY LIKE STRUCTURED, UNSTRUCTURED AND SEMANTIC KNOWLEDGE BY DECIDING OWN WHAT TYPE OF QUERY BY OWN.

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07/09/2026	• Began building the structured data (SQL) layer: designed a sample dataset schema and implemented CSV-to-SQLite loading via pandas. • Added safety validation to the SQL layer, blocking destructive queries (DROP, DELETE, INSERT) before execution — tested against adversarial inputs.
08/09/2026	• Set up Ollama (free, local LLM) for the project, including troubleshooting installation and background-service configuration. • Implemented text-to-SQL generation using the local LLM, converting natural-language questions into validated, executable SQL queries.
09/09/2026	• Built and tested the query router given a question, the system now automatically decides whether to search documents or query structured data, using LLM based tool selection. • Reviewed different tools and frameworks for implementing the knowledge graph component.
10/09/2026	• Wired the router into the main pipeline and confirmed correct routing decisions on real test questions. • Tested different query types to understand how the system should route queries to the relevant data source.
11/09/2026	• Built the final answer-generation layer, combining retrieved evidence (documents and/or SQL results) into a single cited answer. • Implemented a confidence-based refusal mechanism — the system now declines to answer when evidence is insufficient, rather than guessing.
12/09/2026	Holiday (Second Saturday)
13/09/2026	Holiday (Sunday)

Next Week Plan

- Expand the evaluation dataset significantly for testing.

REFERENCE:

https://microsoft.github.io/graphrag/get_started/

https://microsoft.github.io/graphrag/index/default_dataflow/

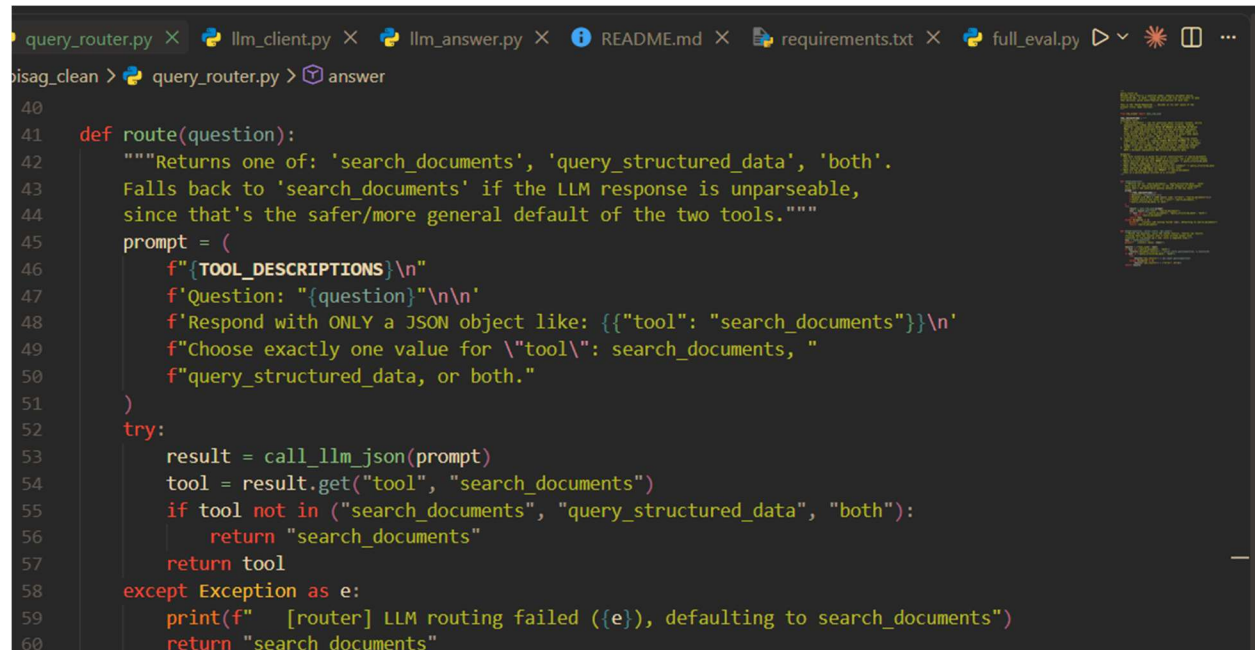
<https://docs.ollama.com/>

<https://www.sqlite.org/docs.html>

<https://github.com/meta-llama/llama3>

<https://docs.python.org/3/library/sqlite3.html>

SCREEN SHOTS:



```
query_router.py X llm_client.py X llm_answer.py X README.md X requirements.txt X full_eval.py
bisag_clean > query_router.py > answer
40
41 def route(question):
42     """Returns one of: 'search_documents', 'query_structured_data', 'both'.
43     Falls back to 'search_documents' if the LLM response is unparseable,
44     since that's the safer/more general default of the two tools."""
45     prompt = (
46         f"{TOOL_DESCRIPTIONS}\n"
47         f'Question: "{question}"\n\n'
48         f'Respond with ONLY a JSON object like: {{"tool": "search_documents"}}\n'
49         f'Choose exactly one value for \"tool\": search_documents, "
50         f"query_structured_data, or both."
51     )
52     try:
53         result = call_llm_json(prompt)
54         tool = result.get("tool", "search_documents")
55         if tool not in ("search_documents", "query_structured_data", "both"):
56             return "search_documents"
57         return tool
58     except Exception as e:
59         print(f" [router] LLM routing failed ({e}), defaulting to search_documents")
60         return "search_documents"
```

Q: Which district has the most PMFBY beneficiaries?
 routed to: query_structured_data -> correct
 SQL: SELECT district FROM scheme_records WHERE scheme_name='PMFBY' ORDER BY beneficiaries_2024 DESC LIMIT 1
 raw result: [('Ahmedabad',)]
 expected value 'Ahmedabad' in RAW SQL result: YES
 expected value 'Ahmedabad' in FINAL answer text: YES
 final answer: According to the evidence, the district with the most PMFBY beneficiaries is Ahmedabad, with the most recent data available in 2024.

Source: [Struct...
 [router] chose: query_structured_data

Q: What is the total budget allocated across all schemes?
 routed to: query_structured_data -> correct
 SQL: SELECT SUM(budget_allocated_inr) FROM scheme_records
 raw result: [(225100000,)]
 expected value '225100000' in RAW SQL result: YES
 expected value '225100000' in FINAL answer text: YES
 final answer: The total budget allocated across all schemes is ₹225,100,000.

Source: [Structured data query: SELECT SUM(budget_allocated_inr) FROM scheme_records]...

